

SYNOPSIS

The container ships *ACX Hibiscus* and *Hyundai Discovery* collided in the eastern approaches to the Singapore Strait on 11 December 2011; visibility at the time was restricted by localised heavy rain. Both vessels' officers of the watch took action to avoid the collision only after they saw the other vessel emerge from the rain shower at close range. This was too late to be effective. Both vessels were severely damaged; nobody was injured and no pollution was reported.

The United Kingdom registered *Hyundai Discovery* was inbound to Singapore and the Panamanian registered *ACX Hibiscus* was outbound from Singapore. The bridge watches on both vessels were being kept by chief officers. *ACX Hibiscus*'s chief officer, unaware of *Hyundai Discovery*'s course and position, turned his vessel to the north to follow its planned route. He did not check that it was safe to turn before he began to alter course. Despite several radio warnings from *Hyundai Discovery*'s chief officer, *ACX Hibiscus*'s chief officer continued to turn his vessel into *Hyundai Discovery*'s path.

The MAIB led a joint investigation into the accident with the Panama Maritime Authority. The investigation found that both vessels were operating in restricted visibility, but neither vessel's bridge watchkeepers had taken the precautions required by the International Regulations for Preventing Collisions at Sea 1972 (as amended). The Panama Maritime Authority investigators concluded that the behaviour of *ACX Hibiscus*'s chief officer was adversely affected by fatigue.

The MAIB's ability to conduct an effective investigation was restricted because access to primary evidence from *ACX Hibiscus*, which was outside the MAIB's jurisdiction, was denied by the vessel's owners. The owners subsequently put pressure on the Panama Maritime Authority not to release critical evidence to the MAIB. As a result, this report has been obliged to focus on the actions that should be taken to avoid rogue vessels, rather than deal with the underlying causes of the accident. The obstructive behaviour of *ACX Hibiscus*'s owners, by influencing a flag state to not comply fully with the International Maritime Organisation's Casualty Investigation Code, has highlighted a significant weakness in the effectiveness of the Code.

Simulations were conducted to consider what action could have been taken by *Hyundai Discovery*'s chief officer when confronted by *ACX Hibiscus* turning towards him. It was concluded that had *Hyundai Discovery*'s chief officer taken action to turn his vessel hard to starboard at the time he began calling *ACX Hibiscus* by VHF radio, *Hyundai Discovery* could have passed 0.4nm ahead of *ACX Hibiscus*.

Hyundai Discovery's managers have delivered training to share the lessons learnt from this accident with the bridge officers in their fleet. They have been recommended to further develop their safety management and training systems to improve inter alia, their watchkeeping officers' understanding of collision avoidance methods and conduct in restricted visibility. The MAIB has also recommended that the Panama Maritime Authority takes appropriate action with the owners of *ACX Hibiscus* to address the underlying causes of ineffective watchkeeping on vessels in their fleet, and to ensure it is compliant with the mandatory standards of the International Maritime Organization's Casualty Investigation Code.

SECTION 1 - FACTUAL INFORMATION

1.1 PARTICULARS OF ACX HIBISCUS AND HYUNDAI DISCOVERY AND ACCIDENT

SHIP PARTICULARS		
Vessel's name	ACX Hibiscus	Hyundai Discovery
Flag	Panama	UK
Classification society	Nipon Kaija Kyokai (NKK)	Lloyd's Register
IMO number	9159141	908576
Type	Container ship	Container ship
Registered owner	ACT Carriers Corp.	Acadia Maritime Ltd.
Manager(s)	Onward Marine Service Co. Ltd.	Zodiac Maritime Agencies Ltd.
Construction	Steel	Steel
Length overall	193.00m	274.67m
Registered length	181.54m	264.95m
Gross tonnage	18,502	64,054
Minimum safe manning	Unknown	Unknown
Authorised cargo	No	No
VOYAGE PARTICULARS		
Port of departure	Singapore	Hong Kong
Port of arrival	Laem Chabang (intended)	Singapore
Type of voyage	International	International
Cargo information	Container	Container
MARINE CASUALTY INFORMATION		
Date and time	11 December 0756 LT	
Type of marine casualty or incident	Serious Marine Casualty	
Location of incident	8nm ENE of the start of the Eastern Singapore Strait Traffic Separation Scheme	
Injuries/fatalities	Nil	Nil
Damage/environmental impact	Structural damage	Structural damage
Ship operation	Container carrier	Container carrier
Voyage segment	On passage	On passage
External & internal environment	Heavy rain	Heavy rain
Persons on board	21	28

1.2 BACKGROUND TO THE INVESTIGATION

This investigation was carried out jointly between the United Kingdom's (UK) Marine Accident Investigation Branch (MAIB) and the Panama Maritime Authority (PMA). It was conducted in accordance with the International Maritime Organization's (IMO) Code of the International Standards and Recommended Practices for a Safety Investigation into a Marine Casualty or Marine Incident (Casualty Investigation Code)¹ (**Annex A**). The PMA agreed that the UK would act as the lead investigating state.

The vessel's owners denied MAIB inspectors access to the Panamanian registered vessel, *ACX Hibiscus*, which at the time was lying outside the MAIB's jurisdiction. Similarly, the MAIB was denied access to *ACX Hibiscus*'s Voyage Data Recorder (VDR). The PMA was not willing to contradict the owner's instructions and did not require that they gave the MAIB access to primary sources of evidence. Consequently, this report has been prepared based on evidence gathered by the MAIB's inspectors, from *Hyundai Discovery*, and a report from the PMA on the circumstances on board *ACX Hibiscus* at the time of the accident.

Timings and navigational data were taken from *Hyundai Discovery*'s VDR, which included a recording of the Automatic Identification System (AIS) data for *ACX Hibiscus*.

1.3 NARRATIVE

1.3.1 Events leading up to the collision

At midday on 8 December 2012, *Hyundai Discovery* departed Hong Kong for the Port of Singapore.

At 0400 on 11 December, *ACX Hibiscus* departed from the Port of Singapore on passage to Laem Chabang in Thailand. *ACX Hibiscus*'s chief officer completed his duties on deck and arrived on the bridge at 0515, where he took over the duty of officer of the watch (OOW) from the third officer. The master was also present on the bridge. At 0700 *ACX Hibiscus* passed Horsburgh Light (**Figure 1**) at a speed of 14.5 knots on an autopilot-controlled heading of 049°. At around the same time, the master left the bridge leaving the chief officer and an Able Bodied Seaman (AB) on watch.

At 0711 *Hyundai Discovery*'s master went to the bridge and talked to the chief officer, who was on watch with a cadet and an AB. *Hyundai Discovery* was making a speed of 20 knots through the water on an autopilot-controlled heading of 203°.

ACX Hibiscus cleared the Singapore Strait Traffic Separation Scheme (TSS) at 0720; the weather was overcast with moderate rain.

At 0721 *Hyundai Discovery*'s master instructed the chief officer to reduce speed to 18 knots in order to arrive at the Singapore pilot station at the correct time. At around this time *Hyundai Discovery* passed through a rain shower and visibility decreased.

¹ IMO Resolution MSC.255(84) entered force in January 2010

At 0730 *ACX Hibiscus* entered a heavy rain shower and the visibility from her bridge reduced accordingly. The chief officer acquired the radar targets of several ships using his Automatic Radar Plotting Aid (ARPA) system, but did not acquire *Hyundai Discovery*.

By 0740 *Hyundai Discovery*'s chief officer had seen and acquired, using ARPA, the radar targets of four vessels that were out-bound from the Singapore Strait TSS. The vessels' positions were fine on his starboard bow. It was reported that heavy rain had reduced the visibility on *Hyundai Discovery*'s port side to approximately 5 miles at around this time.

At about 0740 *Hyundai Discovery*'s cadet looked at the AIS information that was displayed on the Electronic Chart System (ECS) and saw that one of the vessels was *ACX Hibiscus*. The cadet told the chief officer that he thought *ACX Hibiscus* would pass clear down *Hyundai Discovery*'s port side. At 0741 the master returned to the bridge; the chief officer mentioned the recent rainfall to the master. The master instructed the chief officer to inform the on-watch engineer at 0800 that the main engine should be available to manoeuvre by 0900. At 0744 *Hyundai Discovery*'s chief officer used the autopilot control to alter course to starboard to a new heading of 209° (**Figure 2**). During the turn *ACX Hibiscus* was lost from sight.

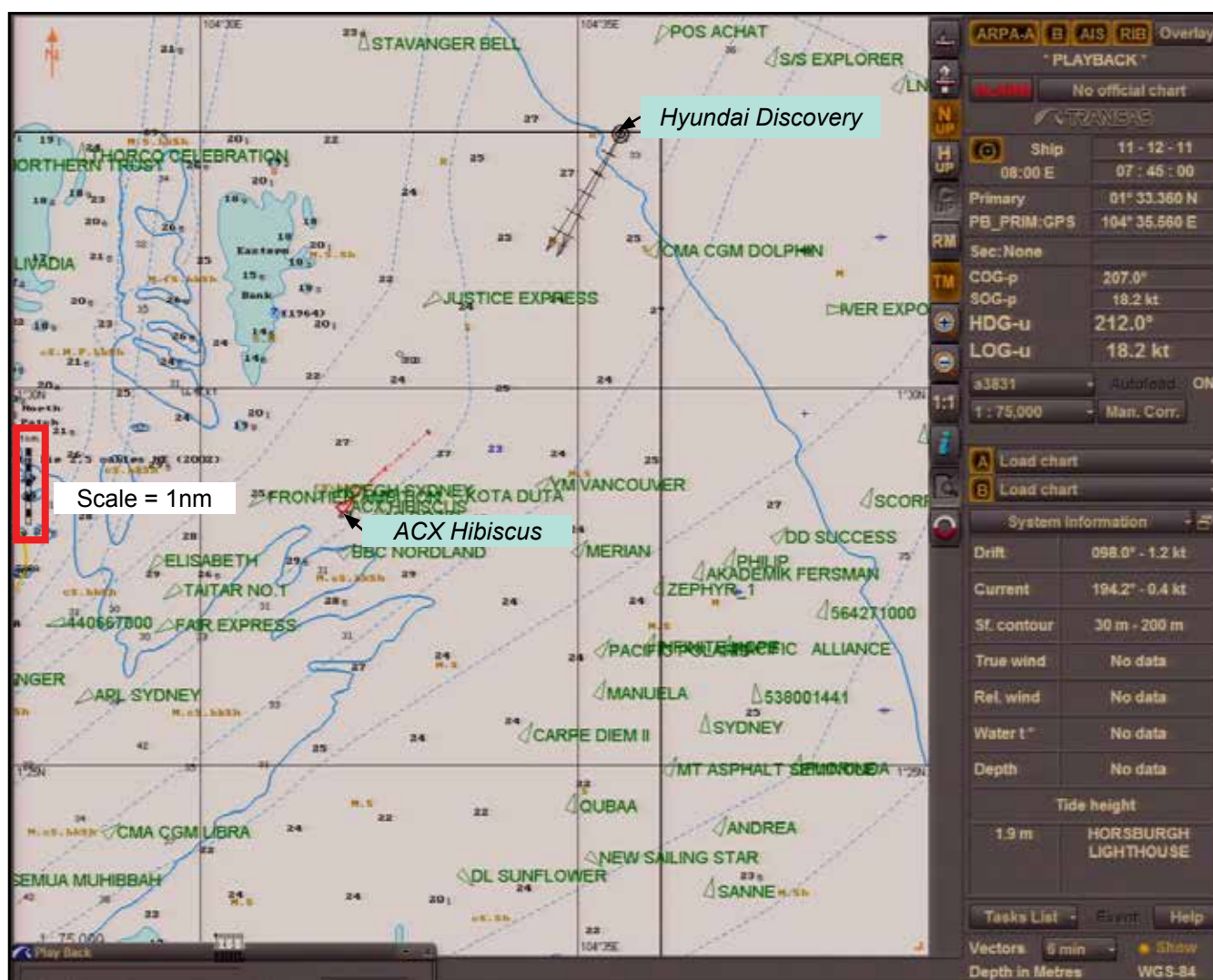


Figure 2: *Hyundai Discovery*'s ECS showing positions of *ACX Hibiscus* and *Hyundai Discovery* at 0744

1.3.2 Watchkeeping in restricted visibility

ACX Hibiscus continued on its heading of 049° at a speed of 14.5 knots. It was reported that, at 0745, the rainfall increased and it was estimated that the visibility from *ACX Hibiscus* reduced to around 2 cables. The rainfall caused the radar picture on *ACX Hibiscus* to become cluttered, and most of the targets that had been acquired were lost from the ARPA radar screen.

At 0747:00 the ARPA radar display on board *Hyundai Discovery* showed that *ACX Hibiscus* was 5nm right ahead of *Hyundai Discovery*. The radar plot indicated that the ships would be passing close, but clear, down each other's port sides (**Figure 3**). The chief officer on *Hyundai Discovery* used the autopilot control to turn to starboard to a heading of 216° to converge onto his next charted course of 229° and to increase his passing distance from *ACX Hibiscus*. At 0748:30, once close to the charted course, the chief officer turned further to starboard onto a heading of 229°.

By 0750:00 *Hyundai Discovery* was steady on a heading of 229°, with *ACX Hibiscus* about 15° on the port bow at a range of 3.5nm (**Figure 4**). The ARPA on *Hyundai Discovery* indicated that *ACX Hibiscus*'s closest point of approach (CPA) would be approximately 7 cables on the port side. *Hyundai Discovery*'s chief officer placed the radar's electronic bearing line (EBL) over *ACX Hibiscus*'s radar target to confirm that its bearing was opening to port as he was expecting.

At 0750 *Hyundai Discovery*'s master returned briefly to the bridge to check the situation, and found that the visibility from his vessel had reduced to about 5 cables in a heavy rain shower. Neither *ACX Hibiscus* nor *Hyundai Discovery* made sound signals for making way in restricted visibility.

1.3.3 Course alteration

ACX Hibiscus's third officer arrived on the bridge at around 0750 ready to take over the watch at 0800. At 0751:30, *ACX Hibiscus*'s chief officer started to alter his vessel's course to port by adjusting the desired heading on the autopilot to turn the ship incrementally onto the next planned track of 350° (**Figure 5**). It was reported that the third officer looked at the radar display and found that radar targets were not seen due to clutter, and that he informed the chief officer about the situation.

At 0752, with about 2.2nm distance between the vessels, *Hyundai Discovery*'s chief officer noticed that the radar trail of *ACX Hibiscus* changed direction towards his vessel. The trail was set to display its motion relative to *Hyundai Discovery* and showed that *ACX Hibiscus* was altering its course to port. The chief officer checked the AIS information on the ECS and confirmed that the vessel turning to port was *ACX Hibiscus*. At 0753:15, and again at 0753:30 (**Figure 6**), *Hyundai Discovery*'s chief officer called *ACX Hibiscus* by VHF radio using the distress and calling channel, channel 16. Once communication was established, both officers changed their VHF radios to a working channel, channel 6.

1.3.4 Action to avoid collision

At 0754:00 *Hyundai Discovery*'s chief officer asked *ACX Hibiscus*'s third officer 'why are you altering course to port side?' The response from *ACX Hibiscus* was unclear, but it was apparent that the third officer said that his vessel was turning to the north. *Hyundai Discovery*'s chief officer continued... 'Do not cross my bow ...the visibility

is zero you must alter course to starboard'. At 0754:20 *Hyundai Discovery*'s chief officer told *ACX Hibiscus*'s third officer to alter course to starboard and create a minimum CPA of 3 cables. He then asked '*the visibility is zero, what are you doing?*'

At 0754:34, *ACX Hibiscus*'s third officer replied that '*we are altering course to the north now*'. The chief officer continued to turn *ACX Hibiscus* to port, passing a heading of 017°.

At 0755:00 *Hyundai Discovery*'s chief officer called *ACX Hibiscus* by VHF radio again. He stated '*go hard to starboard, to starboard... what are you doing?*' *ACX Hibiscus*'s third officer did not reply.

At 0755:13, about 1 minute before the collision, *Hyundai Discovery*'s chief officer ordered the AB to put the helm control into manual steering and he sounded one long blast on the ship's whistle. At 0755:35 *Hyundai Discovery*'s master arrived on the bridge and the chief officer reported the situation with *ACX Hibiscus* to him. *ACX Hibiscus* became visible at about 0755:50 at an estimated range of about 2 cables; the chief officer ordered the helm hard to starboard (**Figure 7**).

At around the same time, *ACX Hibiscus*'s chief officer saw *Hyundai Discovery* become visible on his port bow; he took the steering into manual control, placed the wheel hard to port and set the engine telegraph to 'emergency stop'.

1.3.5 Collision

At 0756:10 *ACX Hibiscus*'s bow collided with *Hyundai Discovery*'s port side wing ballast tank adjacent to No. 2 hold. At the time of the collision *Hyundai Discovery* was heading 229° at a speed of 18 knots; *ACX Hibiscus* was heading 321° at a speed of 14.1 knots (**Figure 8**).

ACX Hibiscus's starboard bow scraped along *Hyundai Discovery*'s port side. Five containers fell from *Hyundai Discovery*'s deck either into the sea, or onto *ACX Hibiscus*'s forward mooring deck. *Hyundai Discovery*'s master placed the rudder hard over to port in an attempt to separate the sterns of the two ships.

ACX Hibiscus's master was alerted by the impact, and went to the bridge; the ship's electrical power failed and propulsion was lost.

Hyundai Discovery listed to port due to the flooding of the empty No.2 wing ballast tank. *Hyundai Discovery*'s master called *ACX Hibiscus* on VHF radio to establish the condition of the other vessel, but the reply from *ACX Hibiscus* was unclear.

At 0800 *Hyundai Discovery*'s master sounded the general alarm and the crew all reported to their muster stations. Several crewmen then assisted in sounding the ship's holds, tanks and void spaces and reported their findings to the chief officer. *Hyundai Discovery*'s chief officer ballasted No.2 starboard wing tank and the list reduced. The master contacted the vessel's managers, Zodiac Maritime Agencies Limited (Zodiac), by satellite telephone and reported the collision to the Designated Person Ashore (DPA). *Hyundai Discovery*'s master reduced to the vessel's manoeuvring speed, which was about 16 knots, and continued on passage to Singapore's outer anchorage.

ACX Hibiscus's master mustered his crew on the bridge and then instructed them to sound the tanks and bilges to check where the hull had been breached. *ACX Hibiscus* was flooded in the fore peak tank and took on a list to starboard. Water also flooded into the forward hold. At 0830 the master advised the Malaysian Coastguard about his ship's damaged status by VHF radio. Following this, the master turned *ACX Hibiscus* to head west at slow speed with the intention of beaching the vessel if necessary to prevent it from sinking. At 0920 the master, satisfied that *ACX Hibiscus's* condition was stable enough to resume passage, navigated his vessel back to Singapore's outer anchorage at slow speed.

1.4 NAVIGATION AND ENVIRONMENTAL CONDITIONS

1.4.1 Approaches to the Port of Singapore from the east

Ships approaching the Port of Singapore from the east and sailing from the port heading east are required to follow the IMO approved TSS (**Figure 1**) that extends from the port to the east of Horsburgh Light. Ships inbound to Singapore, heading south-west, use the northern lane, while the outbound ships follow the opposing southern lane.

1.4.2 Regional climate and weather

The Admiralty Sailing Directions for the Malacca Strait and West Coast of Sumatera (NP44) (**Annex B**) states that '*The mean annual rainfall is abundant... Heavy showers and thunderstorms are responsible for most of the rain*' and '*Visibility is generally good except in thundery showers when visibility may fall to near fog limits*'.

1.4.3 Environmental conditions

The weather and tidal conditions at the time of the accident were:

- Visibility - around 2 cables
- Weather - heavy rain showers
- Wind - Beaufort force 5 from the east-north-east
- Temperature - 29°C
- Tidal flow - west-south-west at 1.7 knots

A picture taken soon after the collision shows the typical conditions at the time (**Figure 9**).

1.5 DAMAGE

1.5.1 Damage to *ACX Hibiscus*

ACX Hibiscus sustained significant damage to its bow section (**Figure 10**). The forward mooring deck, chain locker, fore peak tank and bow thrust room were very heavily damaged. Both anchors were unusable. The collision bulkhead was breached and damage extended into the forward hold.

1.8 REGULATIONS FOR COLLISION PREVENTION

The following rules, from the International Regulations for Preventing Collisions at Sea 1972 (as amended) (COLREGS), are relevant to this accident and are reproduced at **(Annex G)**. These rules are summarised as follows:

- Rule 2 - Responsibility. This rule permits a departure from the collision prevention rules if required to avoid immediate danger.
- Rule 3 - General Definitions. The term 'restricted visibility' includes heavy rainstorms that decrease the available visibility.
- Rule 5 - Lookout. This rule states that, as well as visual lookout, radar and any other means can and should be used when required to assess the risk of collision.
- Rule 6 - Safe Speed. This rule states that a ship should be able to stop in a distance that is suitable for the conditions, taking into account such factors as the visibility, traffic situation and the ship's own manoeuvrability. The rule also requires vessels with radar to assess the limitations of the radars being used, and to consider that not all targets may be displayed owing to interference due to the weather, such as heavy rain.
- Rule 7 - Risk of Collision. This rule requires that all means possible, including radar, should be used to assess if a risk of collision exists as early as possible. This rule does not specifically mention AIS as the last rule amendment predates the introduction of this equipment. However, AIS is considered to be included by the term 'all means'. AIS provides the user with information about other vessels' positions, courses and speeds.
- Rule 8 - Action to Avoid Collision. This rule requires that action to avoid collision is positive, clear, and made in ample time.
- Rule 19 - Conduct of Vessels in Restricted Visibility. This rule confirms that vessels' watchkeepers must have "due regard" for the rules on lookout, safe speed, risk of collision and action to avoid collision while in restricted visibility. The rule applies to vessels out of sight of one another when either vessel is in, or near, an area of restricted visibility. The rule requires that speed should be appropriate to the visibility, and that control of the ship's propulsion should be immediately available. The rule also states that when a risk of collision exists, an alteration of course to port for a vessel forward of the beam should be avoided if possible.
- Rule 35 - Sound Signals in Restricted Visibility. This rule requires that a power-driven vessel making way in restricted visibility should sound the ship's whistle for around 5 seconds every 2 minutes.

1.9 **HYUNDAI DISCOVERY - SIMULATION OF POTENTIAL AVOIDING ACTIONS**

The MAIB carried out a trial, using a full mission bridge simulator, to evaluate what actions *Hyundai Discovery*'s chief officer might have been able to attempt in order to avoid the collision.

Software models representing container vessels of similar sizes and manoeuvring characteristics (**Annex H**) to *Hyundai Discovery* and *ACX Hibiscus* were used in the simulator. Information from *Hyundai Discovery*'s VDR was used to provide positions, courses over the ground, heading, and rate of turn data for both vessels. The simulation was run using the same navigational area, with similar environmental conditions to those at the time of the accident.

The exercise was limited by the fidelity of the software model being used to represent *Hyundai Discovery*. The manoeuvring data was similar, but not identical⁵. The limitations of the simulation are acknowledged and the results are presented to illustrate the credibility of the different options available to avoid collision.

Data from *Hyundai Discovery*'s VDR showed that the time between the chief officer's first indication that *ACX Hibiscus* had altered course, to the time of the collision, was around 4 minutes and 10 seconds. An allowance of 1 minute and 30 seconds was made for the time needed to assimilate the situation and determine that the risk of collision existed. This was comparable with the time that elapsed before the chief officer on *Hyundai Discovery* made his first VHF radio call to *ACX Hibiscus*.

The simulation considered alterations of course to starboard, port and an emergency stop. Action to avoid collision was taken:

- 1 minute and 30 seconds after identifying that *ACX Hibiscus* had altered course (2 minutes 40 seconds before collision)
- 2 minutes after identifying that *ACX Hibiscus* had altered course (2 minutes 10 seconds before collision)
- 3 minutes after identifying that *ACX Hibiscus* had altered course (1 minute 10 seconds before collision).

ACX Hibiscus was assumed to maintain the same courses and speeds as during the accident.

⁵ The ship's particulars and manoeuvring data used in the simulation are shown at **Annex H**.

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 THE COLLISION

ACX Hibiscus's chief officer altered his vessel's course to port, in restricted visibility, into the path of *Hyundai Discovery*. The chief officer was told by the third officer that some radar targets could not be seen because of clutter. He was further alerted by the VHF radio calls from *Hyundai Discovery's* chief officer asking him to alter course back to starboard. Despite these warnings, the chief officer on *ACX Hibiscus* continued to turn his vessel.

Hyundai Discovery's chief officer noticed *ACX Hibiscus* alter course soon after it began to turn, about 4 minutes before the collision occurred. This gave him little opportunity to assimilate this unexpected manoeuvre and take avoiding action.

2.3 RESTRICTED VISIBILITY

Both vessels had either entered, or were near to, areas of restricted visibility caused by heavy rain before the collision; *ACX Hibiscus* for at least 25 minutes and *Hyundai Discovery* for around 15 minutes. Immediately prior to the collision, the visibility had significantly reduced in the heavy rain and was reported to be as little as 2 cables. Consequently, both vessels were considered to be affected by restricted visibility as defined by the COLREGS. Neither vessel was in sight of the other, and consequently Rule 19 of the COLREGS - Conduct of Vessels in Restricted Visibility - applied to both *ACX Hibiscus* and *Hyundai Discovery*.

Rule 19 requires that vessels proceed at a safe speed, and that their engines are ready for immediate manoeuvring. This rule also states that all available means are used to establish whether a risk of collision exists. In addition, Rule 35 required that the whistle be sounded.

ACX Hibiscus's chief officer did not call the master to report the restricted visibility. *Hyundai Discovery's* master was aware of the reduced visibility as he had been on the bridge intermittently prior to the collision. The master reduced speed, but only to arrive at the pilot station on time. Neither vessel's bridge teams reduced to manoeuvring speed, enhanced their ability to keep a lookout (either by eye or radar), transferred from autopilot to manual steering, or made the required sound signal.

Both vessels' bridge teams would have been much better prepared to respond to the traffic situation which they faced had they taken effective action to comply with the COLREGS.

cross the paths of the south-west bound traffic once her course had been altered to the north. This should have emphasised the importance of checking that it was safe to turn to port.

There were no navigational restrictions or other pressures reported which obliged the chief officer to make the course alteration at that precise moment. It is possible that the chief officer wanted the ship to be steady on its new course before he handed over the watch to the third officer. However, this did not justify the risk of making a course alteration without adequately checking that it was safe to do so.

It is difficult to imagine that an experienced chief officer would have deliberately initiated an alteration of course into *Hyundai Discovery*'s path. It is therefore concluded that the chief officer was not aware of *Hyundai Discovery*'s position and course, or the hazardous situation that was developing.

The chief officer had insufficient situational awareness due to his ineffective use of the navigational aids that were available to him. Notwithstanding the potential effect of fatigue on the chief officer, the standard of watchkeeping on *ACX Hibiscus* at the time of the accident was far below the standard that should be expected of a qualified watchkeeper. Due to the limited evidence available to the MAIB, it has not been possible to examine the underlying reasons for these failings.

2.4.3 Continuation of the turn to port

The third officer arrived on *ACX Hibiscus*'s bridge to take over the watch from the chief officer as the vessel started to turn. When he responded to the initial VHF radio call from *Hyundai Discovery*'s chief officer, potentially without appreciating the significance of the message, the third officer's response to the question '*why are you altering course to port?*' was that *ACX Hibiscus* was '*turning to the north*'. It is considered likely that in this response the third officer was relaying information passed to him by the chief officer.

It was reported that the third officer looked at the radar display and found that radar targets were not seen due to clutter, and that he informed the chief officer about the situation. Although it was not clear from the information provided to the MAIB, this conclusion could only have been drawn by the third officer if he had adjusted the clutter control and subsequently observed radar targets. These probably included *Hyundai Discovery*. However, the chief officer continued to turn *ACX Hibiscus* to port.

The VHF radio calls from *Hyundai Discovery* did not alert *ACX Hibiscus*'s chief officer to the imminent danger that he faced, and did not prompt him to consider the situation further or stop him from continuing his vessel's turn to port.

The reason why the information, and manner, of the report from the third officer and the VHF radio call from *Hyundai Discovery* did not convince *ACX Hibiscus*'s chief officer to reconsider his vessel's manoeuvre was not examined in the PMA's investigation report.

2.4.4 Response to seeing *Hyundai Discovery*

When *Hyundai Discovery* appeared out of the heavy rain at a range of around 2 cables, *ACX Hibiscus*'s chief officer recognised that there was an imminent risk of collision. The chief officer put *ACX Hibiscus*'s rudder hard to port, to increase the rate of turn, and stopped the propulsion in an attempt to avoid collision. These actions were too late to be effective.

The MAIB did not have access to *ACX Hibiscus*'s VDR, and it was therefore not possible to determine exactly when avoiding action was taken. The ship's turning circle data was not provided. Consequently, it was not possible to simulate alternative responses, such as steering to starboard and passing round *Hyundai Discovery*'s stern, which might have been more effective.

2.4.5 Passage planning

The plan for *ACX Hibiscus* to make passage from Singapore to Laem Chabang did not consider the likelihood of heavy rain restricting visibility, nor did the plan consider the risks of altering course to the north across the south-west bound traffic heading for the Singapore Strait TSS. The master had not signed the passage plan, so the planning process was incomplete, and it is considered very likely that it had not been discussed with the watchkeepers prior to departure from Singapore.

Had the passage planning process considered the possibility of heavy rain reducing visibility while the vessel was crossing a busy traffic lane, the master could have identified that additional resources were needed to mitigate the risks during the alteration of course to the north. The master could have remained on the bridge to assist the watch officers, or increased the watch manning.

The master's lack of oversight of the passage plan meant that appropriate bridge resources were not provided during an entirely foreseeable combination of hazardous circumstances.

2.4.6 Actions following collision

ACX Hibiscus's master was alerted to the collision by the impact. His subsequent actions and those of the crew were not recorded or analysed in the PMA's investigation report.

2.4.7 Summary

From the limited information provided, it is considered most likely that the chief officer's performance was adversely affected by fatigue; that the standards of watchkeeping applied on *ACX Hibiscus* at the time of the accident were extremely poor; and that the master's oversight of both the passage plan and his officers' performance was ineffective.

It was not possible to draw any more detailed conclusions from the information provided by the PMA.

2.5 HYUNDAI DISCOVERY

2.5.1 Fatigue

There was no evidence that *Hyundai Discovery*'s chief officer was fatigued at the time of the accident, and fatigue is not considered to have had any effect on his actions.

2.5.2 Actions of *Hyundai Discovery*'s chief officer

The chief officer identified that *ACX Hibiscus* was turning to port within a minute of the turn starting; he then had around 4 minutes to react before the two vessels collided. The turn was executed quickly and was a significant alteration of course. There was no warning that *ACX Hibiscus* was about to alter course, and the chief officer had a very limited time to respond to what was a surprising and hazardous situation. This was compounded by the effect of the restricted visibility; that neither vessel was making sound signals; and that *Hyundai Discovery*'s chief officer had to rely on his radar and AIS to determine what was happening.

The chief officer's response to the situation was to use the VHF radio to alert *ACX Hibiscus*'s chief officer, question his actions, and ask him to reverse the turn to port and turn to starboard instead. However, the chief officer was unable to alert *ACX Hibiscus*'s chief officer to the danger he was in, or persuade him to change his actions.

After *Hyundai Discovery*'s chief officer's final VHF radio transmission to *ACX Hibiscus*, there was just over a minute left until the collision occurred. By this time, the actions of *Hyundai Discovery* alone were very unlikely to be sufficient to avoid collision (see Section 1.9). It was only during the final minute that *Hyundai Discovery*'s chief officer switched the helm control to manual steering and sounded the whistle. The helm was put hard to starboard just 20 seconds before the collision.

The chief officer's decision to use the VHF radio to try and persuade *ACX Hibiscus*'s chief officer to reverse his actions was an understandable reaction in the circumstances, as the most effective way of avoiding the collision was for *ACX Hibiscus* to alter course to starboard. If the bridge team on *ACX Hibiscus* had been more alert, it might have been successful. However, the VHF calls were contrary to the instructions in Zodiac's SMS, offered no guarantee of success, and wasted valuable time. Time can appear to pass very quickly in a stressful situation, and every minute is vital when manoeuvring a vessel the size of *Hyundai Discovery*.

This accident should be an important reminder to all bridge watchkeeping officers, as required by COLREGs Rule 8, that action to avoid collision should be substantial and taken in good time.

2.5.3 Actions available to the chief officer

Hyundai Discovery's chief officer was quick to notice *ACX Hibiscus*'s turn to port. He called *ACX Hibiscus* by VHF radio at 0753:15, and again at 0753:30, around 2 minutes 40 seconds before the collision. This time was considered to mark the point at which the chief officer had assessed the situation and decided what action to take.

Results obtained from the ship simulator trials were used to determine what actions would have been the most effective, and the latest time at which action could have been taken to avoid the collision. The emergency actions available to the chief officer were to turn hard to starboard, turn hard to port or initiate a crash stop. There were no navigational hazards to restrict the range of emergency options, and each could have been completed without causing a collision with a vessel other than ACX *Hibiscus*.

- Crash stop

The simulator results showed that, due to the time lag built into the ship's engine control system, a crash stop would not have altered *Hyundai Discovery*'s speed significantly. This option would not have prevented the collision.

Had the engine been ready to go astern immediately (if the engine was operating at a speed below which it could be quickly stopped or reversed without causing damage), it is possible that this option could have been more effective.

- Rudder hard to port

It could be argued that an alteration to port by *Hyundai Discovery* would have been permissible under Rule 2 of the COLREGS. It was estimated from the simulation that if the chief officer placed the rudder hard to port at 0753:30 (2 minutes 40 seconds) before the collision, *Hyundai Discovery* would have passed 3.5 cables ahead of ACX *Hibiscus*. Delaying by another 30 seconds would have reduced the bow crossing range to 1.5 cables.

This option would have been counter-intuitive to the chief officer as it would have been contrary to Rule 19 of the COLREGS and would possibly have placed the ship in greater danger if ACX *Hibiscus* had turned to starboard as *Hyundai Discovery*'s chief officer requested. Also, there was greater traffic density in this direction, which could have created additional risks.

- Rudder hard to starboard

Turning *Hyundai Discovery* to starboard complied with all the applicable COLREGs, took the ship away from the majority of traffic, and separated it from any further actions taken by ACX *Hibiscus*. Had a hard turn to starboard been initiated 2 minutes and 40 seconds before the collision, it was estimated that *Hyundai Discovery* would have passed 4 cables ahead of ACX *Hibiscus*. A delay of a further 30 seconds would have halved this range to 2 cables.

A turn to starboard was considered to be the most effective action to avoid the collision. It generated the greatest passing distance, ensured that *Hyundai Discovery* was not affected by the actions of ACX *Hibiscus*, and complied with the COLREGS.

If the chief officer had turned hard to starboard at the same time as he called ACX *Hibiscus* by VHF radio, it is likely that *Hyundai Discovery* would have passed around 4 cables ahead of ACX *Hibiscus*. *Hyundai Discovery* was not constrained by traffic or navigational hazards, and the chief officer could have continued to turn the vessel short round or, once ahead of ACX *Hibiscus*, turned to port and regained his original track.

2.5.4 Knowledge of emergency actions

Hyundai Discovery's turning circles at various speeds and loading conditions, along with a diagram showing the time taken to achieve a crash stop, were posted on the bridge.

Although the chief officer was fully aware of this information, he had not translated this knowledge into a practical understanding of when to apply it. In the circumstances of this accident, the chief officer needed to know the effect of putting the rudder hard over, and be confident in taking such bold action. There was no time to delay, and had the chief officer properly understood this, his decision making process might have been different.

The master's orders allowed officers to take emergency actions if required. However, unless officers feel confident about the likely outcome of making an emergency manoeuvre, they might be reticent to make such a manoeuvre. Similar emergency manoeuvres might also be required to avoid a grounding, or following a man overboard incident.

Zodiac's bridge officers would benefit from training and practice in how best to manoeuvre their ships in close quarter situations, to take action to avoid collision, or other emergency situations. This training could be achieved through simulator or familiarisation training. However, the simplest and most effective method could be to set aside training periods where OOWs can practise emergency manoeuvres with their vessels in controlled and supervised conditions while at sea.

2.5.5 Use of VHF for collision avoidance

Both Zodiac's and the master's instructions stated that VHF radio should not be used for collision avoidance. The instructions stated that VHF radio conversations can introduce confusion and that positive identification of the vessel is not guaranteed.

Although AIS had provided the chief officer with a positive identification of *ACX Hibiscus*, as discussed earlier in this report, the VHF radio call did not have the desired effect on *ACX Hibiscus's* chief officer.

While it is likely that *ACX Hibiscus's* chief officer was adversely affected by fatigue, there are many other potential reasons why an OOW may not respond to a VHF call. These might include: other, more pressing work; faulty equipment; not understanding the language used during transmission; or even that the bridge is not manned.

It is acknowledged that VHF radio calls are routinely used by OOWs to advise other vessels of their intentions and to request clarification of another vessel's movements. However, this method cannot be relied on for collision avoidance, and OOWs must not allow the use of VHF radio to delay them from taking action.

The lessons learnt from this accident should serve as a timely reminder of the risks of using VHF radio for collision avoidance, as highlighted in MGN 324 (M+F).

2.5.6 AIS information and presentation

The primary display of AIS information was on the unit located close to *Hyundai Discovery's* chart table. This unit provided either a list of the vessels in the vicinity, or a visual representation of the traffic situation. The unit's screen was small, and

SECTION 4 - CONCLUSIONS

4.1 SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT WHICH HAVE RESULTED IN RECOMMENDATIONS

1. *ACX Hibiscus's* chief officer altered course to port, in restricted visibility, into *Hyundai Discovery's* path. Despite warnings from his own third officer and the VHF radio, the chief officer on *ACX Hibiscus* continued to turn his vessel. [2.2]
2. *Hyundai Discovery's* chief officer noticed that *ACX Hibiscus* began to turn about 4 minutes before the collision occurred. This gave him little opportunity to assimilate this unexpected manoeuvre and take avoiding action. [2.2]
3. Neither vessel's bridge teams complied fully with the applicable COLREGS for restricted visibility. Both vessels' bridge teams would have been much better prepared to respond to the traffic situation which they faced had they taken effective action to comply with the COLREGS. [2.3]
4. *ACX Hibiscus's* chief officer had worked for 19 of the 24 hours before the accident. The PMA's investigation report concluded that the chief officer's performance was adversely affected by fatigue, and that this was causal to the accident. [2.4.1]
5. Neither *ACX Hibiscus's* master nor the chief officer recognised that the effects of fatigue made the chief officer unsuited to keeping a watch in an area affected by high traffic density and restricted visibility. The watchkeeping routine on *ACX Hibiscus* should have been adapted to avoid taking such a risk. [2.4.1]
6. Notwithstanding *ACX Hibiscus's* chief officer's potential fatigue, the standard of watchkeeping on the vessel at the time of the accident was far below that which should be expected of a qualified watchkeeper. [2.4.2]
7. The reasons why *ACX Hibiscus's* chief officer started to turn his vessel to port without checking that it was safe to turn across the opposite traffic route, and continued to turn despite warnings, were not examined in the PMA's investigation report. [2.4.2, 2.4.3, 3.3]
8. *ACX Hibiscus's* chief officer took action to avoid collision only when *Hyundai Discovery* became visible. By that time, it was too late to avoid a collision. [2.4.4]
9. *ACX Hibiscus's* master's lack of oversight of the passage plan meant that appropriate bridge resources were not provided during an entirely foreseeable combination of hazardous circumstances. [2.4.5]
10. This accident demonstrates that using VHF radio calls to try and avoid a collision offers no guarantee of success, and can waste valuable time. Action to avoid collision should be substantial and made in good time. [2.5.2]

4.2 OTHER SAFETY ISSUES IDENTIFIED DURING THE INVESTIGATION ALSO LEADING TO RECOMMENDATIONS

1. That *ACX Hibiscus*'s owners were able to apply pressure to PMA, which resulted in much key evidence being withheld from the lead investigating state, has highlighted a fundamental weakness in that administration's application of the IMO Casualty Investigation Code. [3.4]

4.3 SAFETY ISSUES IDENTIFIED DURING THE INVESTIGATION WHICH HAVE BEEN ADDRESSED OR HAVE NOT RESULTED IN RECOMMENDATIONS

1. Simulations undertaken by the MAIB showed that, when *ACX Hibiscus* turned to port, the most effective action that *Hyundai Discovery*'s chief officer could have taken was to turn his vessel to starboard. Had he done this at around the same time as he called *ACX Hibiscus* by VHF radio, it is likely that *Hyundai Discovery* would have passed around 4 cables ahead of the other vessel. [2.5.3]
2. Zodiac's bridge officers would benefit from training and practice in how best to manoeuvre their ships in close quarter situations to take action to avoid collision, or other emergency situations. [2.5.4]
3. The lessons learnt from this accident should serve as a timely reminder of the risks of using VHF radio for collision avoidance. [2.5.5]
4. Following the accident, neither vessels' master took appropriate action to offer assistance to the other vessel as required by UNCLOS. This accident serves as a reminder of a master's obligation to offer assistance to any other vessels that his vessel is in collision with. [2.5.7]
5. The PMA, despite agreeing that the UK would act as the lead investigating state, allowed the owners of *ACX Hibiscus* to deny the MAIB access to critical evidence. [3.2]
6. Due to a lack of access to critical evidence, this report has been obliged to focus on the actions that should be taken to avoid rogue vessels, rather than deal with the underlying causes of the accident. [3.3]